

Pathfinder as a state machine

Pathfinder, github.com/vd1/pathfinder

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One pair of papers is one machine. The campaign is a set of these machines plus a scheduler with a purse. Everything below is what the code does; the builder prompt points here for the shape to re-implement.

1 Global state of a pair

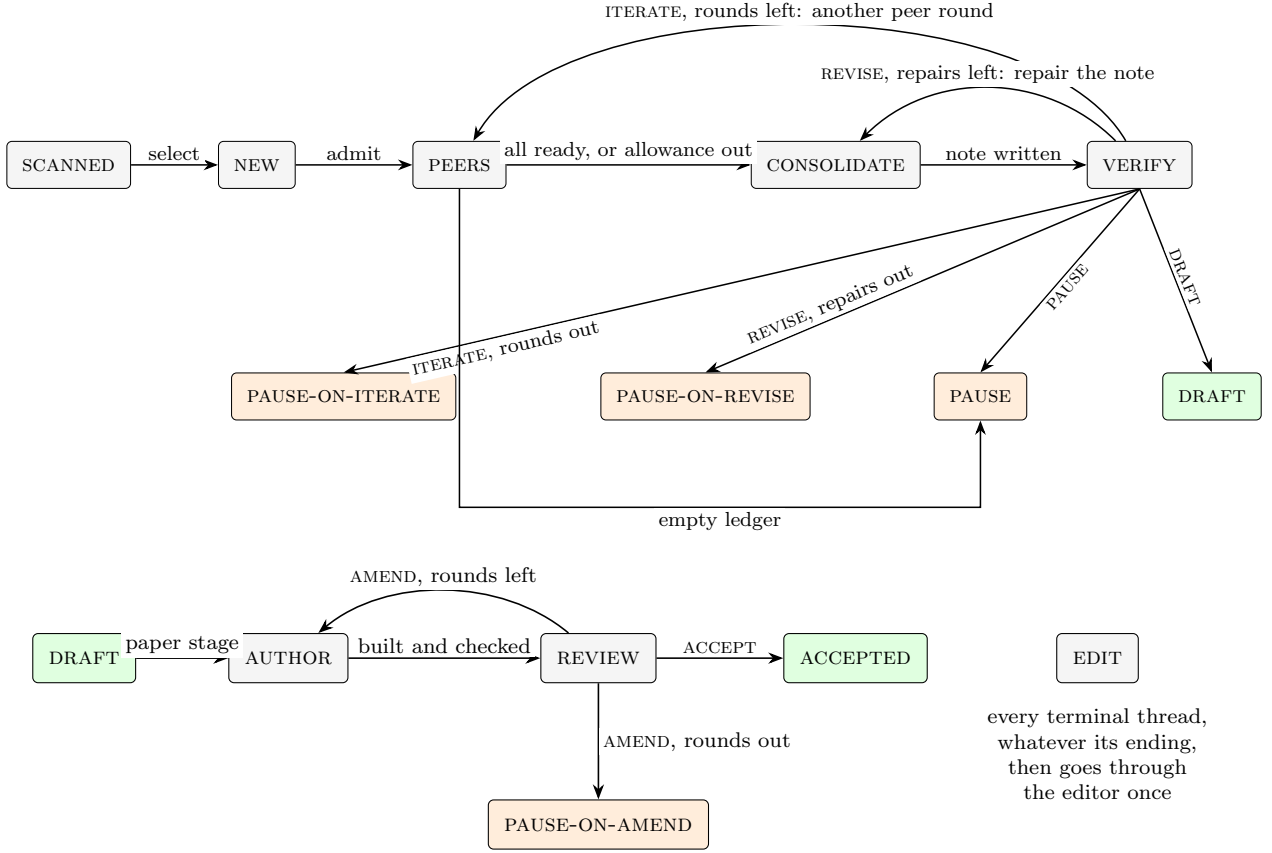
Position. `stage`, where the machine is; `status`, one of `running`, `stopped`, `BLOCKED` or a terminal word; the counters `round` (research rounds used), `repairs` (note repairs used), the paper's `round` and the editor's `attempt`.

Memory, all files under `threads/<pair>/`:

- `inputs/`: the two papers, flattened $\text{\TeX}$ or PDF text, and their metadata.
- `ledger.jsonl`: append-only attributed entries of kind `idea`, `finding`, `objection`, `correction`, `intention`, `ready` or `review`; the peers' shared memory and the verifier's evidence.
- `ada/`, `emmy/`, ...: each peer's derivations, scripts and checks.
- `<pair>.tex`: the consolidated note; `<pair>.verdict.json`: every verdict with the digest of the note it judged.
- `edited/`: the readable note (`note.tex`, `references.bib`, `note.pdf`).
- `paper/`: the paper, its search record and every review with the digest of the paper it judged.
- `status.json`, `paper/paper.json`, `edited/edit.json`: the position.

Campaign level. `scan.jsonl`, `shortlist.json`, `receipts.jsonl` (the only spend figure), the markers `stop.json` and `health.json`, and one pid lock per pair.

2 States



Terminal words of the research thread: DRAFT, PAUSE, PAUSE-ON-ITERATE, PAUSE-ON-REVISE. Terminal words of the paper: ACCEPTED, PAUSE-ON-AMEND. Every PAUSE-ON-X means “the loop named X ran out of its budget with the judge still asking”; a human can raise the budget and resume from the recorded stage. The editor runs once for every terminal thread, whatever its ending. **stopped** and **BLOCKED** are not endings: **stopped** is a drain checkpoint (a stop marker, a transport failure, a kill) and resumes at the recorded stage; **BLOCKED** is an unreadable decision or a missing output and waits for **reconcile**.

3 Transitions

Each transition is one or more model calls. “Sees” is the part of the global state the agent can read; “writes” is what it may change. Nothing else is visible or writable, which is what makes the record auditable.

from → to	agent, prompt, tools	sees	writes	budget
corpus → SCANNED	judge, scan.md , no tools	Q and P titles and abstracts, or full text	one row of scan.jsonl	one call, retried once if unparseable
SCANNED → NEW	none (select)	all scan rows	shortlist.json	none
NEW → PEERS	none (runner)	shortlist, statuses, locks, receipts, markers	lock, status.json	seats; guard: spend plus in-flight estimate under budget_usd , else the stop marker
PEERS → PEERS	every peer concurrently, peer.md , tools and web search	inputs, ledger, own and partners' directories, the scan's connexion and scores, this call's allowance	ledger entries through the helper, own directory	peer_calls per peer and peer_seconds shared, per round; a peer whose ready stands waits instead of calling
PEERS → CONSOLIDATE	none	ledger readiness	status.json	when every peer is ready or the allowance is out; an empty ledger goes to PAUSE
CONSOLIDATE → VERIFY	first peer, consolidate.md , tools	ledger, all directories, the prior note and the verifier's review if any	<pair>.tex	consolidate_seconds ; rerun once unless the note exists
VERIFY → next	verifier, verify.md , no tools	inputs, ledger, note, all inline	verdict file, a review entry in the ledger, status.json	verify_seconds ; rerun once on an empty reply
VERIFY → PEERS	verdict ITERATE		round +1	rounds (4); at the cap PAUSE-ON-ITERATE
VERIFY → CONSOLIDATE	verdict REVISE	the consolidator also sees the corrections	repairs +1	repairs (1); at the cap PAUSE-ON-REVISE
DRAFT → REVIEW	author, author.md , tools and web search	everything in the thread, the web, the previous review's findings if any	paper/	paper_seconds per call
REVIEW → next	reviewer, review.md , no tools	paper, bib, the pipeline's reference checks, the search record, note, ledger, inputs, all inline	paper/review.json , paper.json	review_seconds ; paper_rounds (3); at the cap PAUSE-ON-AMEND
terminal → EDIT	editor, editor.md , tools, no search	inputs and metadata, note, verdicts, ledger, all directories, the build and reference checks on retry	edited/	edit_seconds ; two attempts, then blocked

Between transitions the pipeline itself, with no model, does the checks that gate the next state: it builds the $\text{\LaTeX}$, verifies citations against the bibliography and arXiv, records digests, and writes the position. The verifier's and reviewer's judgements are the only decisions; every other arrow is mechanical.

4 Budgets, and what exhausting them means

budget	scope	when exhausted
<code>peer_calls</code> , <code>peer_seconds</code>	one peer round	consolidate on the ledger as it stands
<code>rounds</code>	the ITERATE loop	PAUSE-ON-ITERATE
<code>repairs</code>	the REVISE loop	PAUSE-ON-REVISE
<code>paper_rounds</code>	the AMEND loop	PAUSE-ON-AMEND
editor attempts	the build and reference checks	blocked
<code>budget_usd</code>	the campaign, at admission	stop marker; threads in flight finish; NEW pairs w

Every model call appends a receipt, so the campaign budget is checked against the sum of receipts, never against an estimate alone.

5 Interrupts

- A stop marker (operator, guard, or first Ctrl-C) stops admission; calls in flight land and write their checkpoints; the runner exits; the next start resumes each pair at its recorded stage.
- No session within the grace time is a transport failure: no receipt, the pair is **stopped**, two in a row set the health marker and pause admission until a probe succeeds.
- **reconcile** reads the position of one pair and names the one safe action: start, resume peers, run consolidate, run verify, or nothing; **--apply** performs it.